The Rise of Multimodal Learning: Vision and Language Meet

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University of Colorado Boulder
Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture
  - Motivation: when vision asks not just “what”, but also “where”
  - Redesigning neural networks to predict densely, rather than just classify
  - Freeing dense prediction networks from fixed input sizes
  - Pivotal insights: pretrained features + design tricks to boost spatial precision
  - Programming tutorial

- Assignments (Canvas)
  - Lab assignment 2 due Friday (answers to common questions on assignment page)
  - Problem set 2 due in a week

- Questions? (can also post these to Piazza or attend office hours)

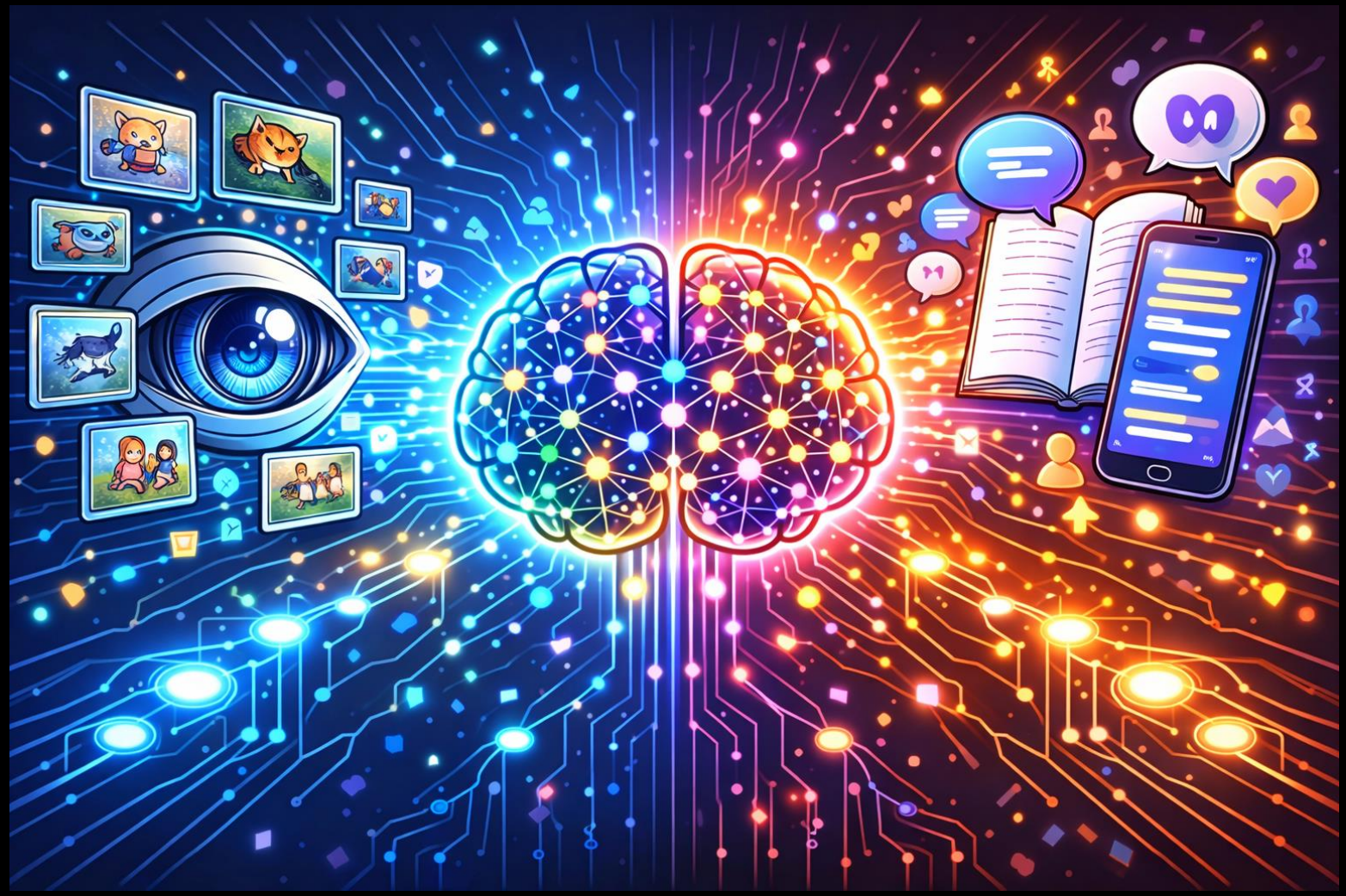
Today's Topics

- Isolated successes: CNNs for vision and RNNs for language
- The next challenge: describing visual scenes with natural language
- Early vision-language pioneers: connecting pixels and words
- Programming tutorial

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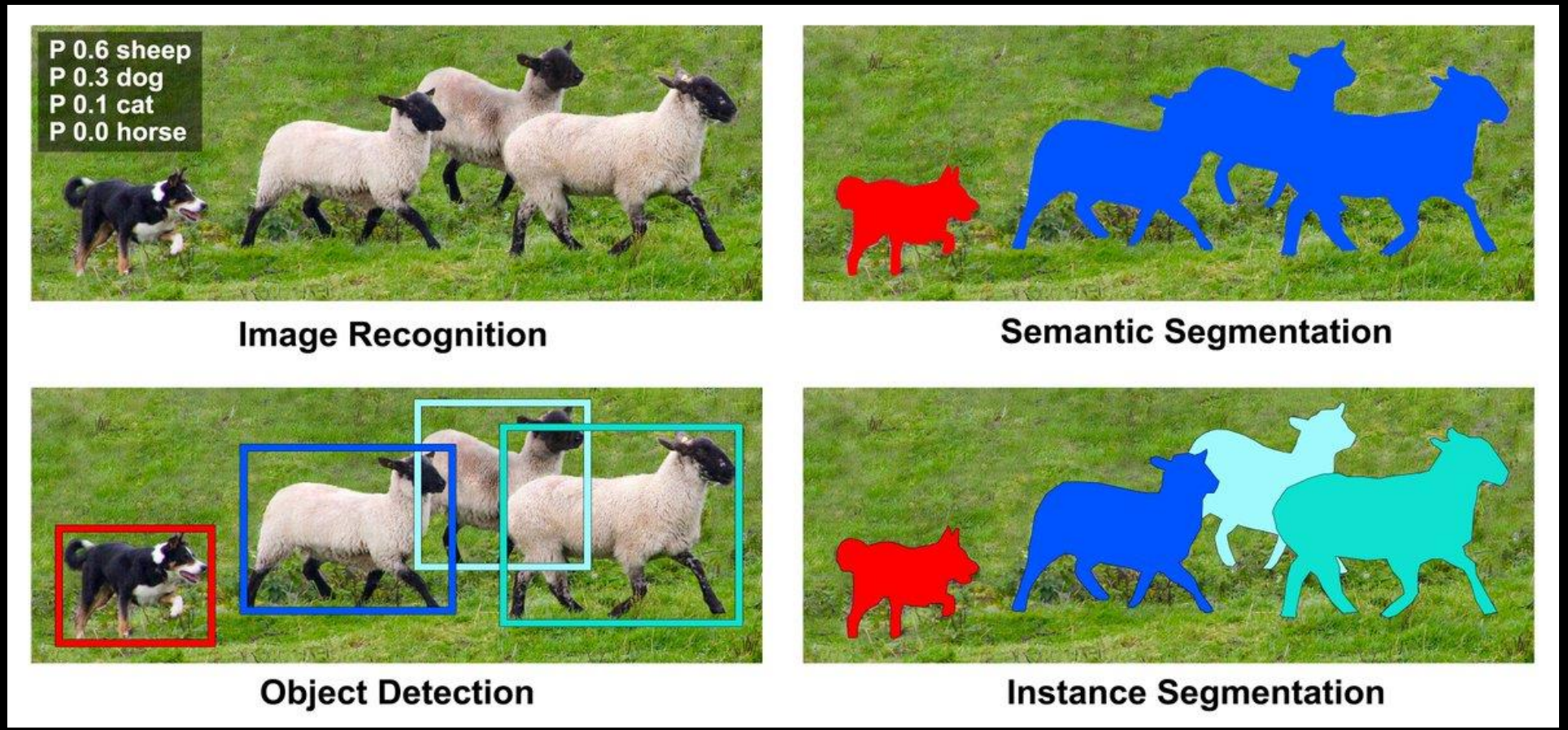
Neural Networks Solidified As Dominant Solution for Vision and Language Problems… In Isolation



CNNs Dominated "Bucketed" Vision Tasks (and Successes Translated to Real-World Applications!)

ResNet in 2015

FCN in 2015



Faster R-CNN in 2015

Mask R-CNN in 2015

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

RNN Dominated Language Tasks; e.g., Machine Translation by Seq2Seq

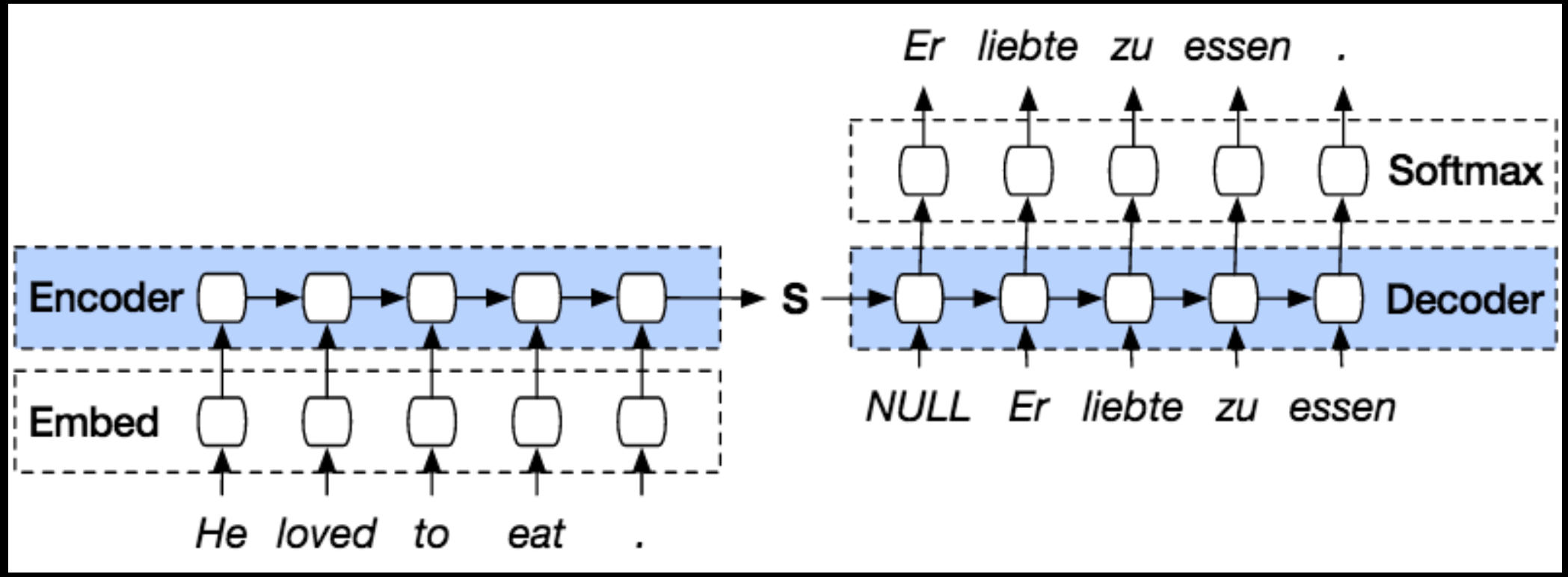


Image source: https://smerity.com/articles/2016/google_nmt_arch.html
seq2seq: Sutskever et al. Sequence to Sequence Learning with Neural Networks. Neurips 2014.

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Multimodal Tasks: Uses 2+ Modalities

e.g., computer vision + natural language processing tasks (open-ended responses!)



Caption:

A bunch of small light brown mushrooms in a green field.

Answer Visual Question:

Q: Is it edible or poisonous?

A: Poisonous

Many Motivating Applications; e.g.,

Visual Interpretation for Blind People

Seeing AI

Lookout by Google

envision

TapTapSee

ORCAM

be my eyes

Personal Assistant



Describing and Responding to Images with “Personality”



Standard captioning output: A plate with a sandwich and salad on it.

Our model with different personality traits (215 possible traits, not all shown here):

Sweet	That is a lovely sandwich.
Dramatic	This sandwich looks so delicious! My goodness!
Anxious	I’m afraid this might make me sick if I eat it.
Sympathetic	I feel so bad for that carrot, about to be consumed.
Arrogant	I make better food than this
Optimistic	It will taste positively wonderful!
Money-minded	I would totally pay $100 for this plate.

Shuster et al. Engaging Image Captioning via Personality. 2019

Pioneers of Multimodal Learning Movement



Larry Zitnick
Formerly, Microsoft Researcher
Meta Research Director

(arXiv, 2015)

Microsoft COCO Captions: Data Collection and Evaluation Server

Xinlei Chen, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam
Saurabh Gupta, Piotr Dollár, C. Lawrence Zitnick



Devi Parikh
Formerly, Professor & Meta Research Director
Co-Founder/CEO of Yutori

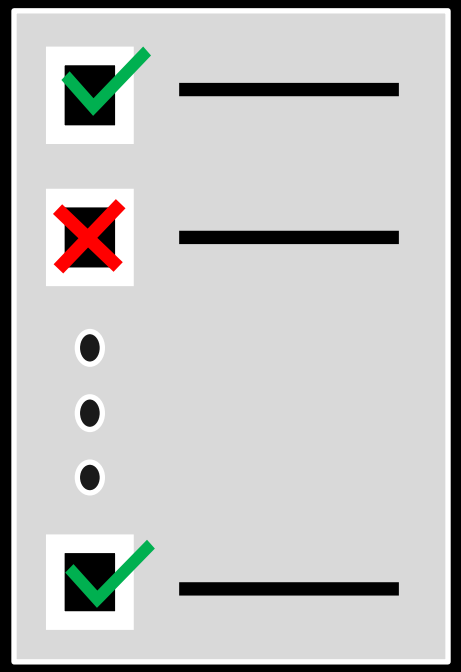
(ICCV, 2015)

VQA: Visual Question Answering

www.visualqa.org

Aishwarya Agrawal*, Jiasen Lu*, Stanislaw Antol*,
Margaret Mitchell, C. Lawrence Zitnick, Dhruv Batra, Devi Parikh

Approach: Large-Scale Dataset Challenges



Ingredients for each challenge:

1. Examples with target results for training, validation, & testing
2. Metric for assessing the similarity between each model prediction and the target result
3. Evaluation servers with leaderboards for standardizing evaluation and enabling world-wide competition
4. New challenges for the community to tackle, evidenced by dataset analysis and model benchmarking

Image Captioning

- 1,026,459 captions collected for 164,062 images from AMT workers
- How long do you think data collection took?
  - ~8,500 hours, equivalent to ~4 years for one person (40 hrs per week, 52 weeks per year)
- How much do you think it cost?
  - ~$128k (assumes 30 sec/caption and $15/hour; would yield for one person $32,000 per year)

Chen et al. Microsoft COCO Captions: Data Collection and Evaluation Server. arXiv 2015.

VQA

- 614,163 questions collected for 204,721 images from AMT workers
- 1,463,510 answers collected for 145,351 images from AMT workers

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Caption Generation:



Instructions:

- Describe all the important parts of the scene.
- Do not start the sentences with "There is".
- Do not describe unimportant details.
- Do not describe things that might have happened in the future or past.
- Do not describe what a person might say.
- Do not give people proper names.
- The sentence should contain at least 8 words.

Please describe the image:

Enter description here

prev next

5 captions collected from
5 crowdworkers

Chen et al. Microsoft COCO Captions: Data Collection and Evaluation Server. arXiv 2015.

Question Generation:

Stump a smart robot! Ask a question about this scene that a human can answer, but a smart robot probably can't!

Updated instructions: Please read carefully

Hide Show

We have built a smart robot. It understands a lot about scenes. It can recognize and name all the objects, it knows where the objects are, it can recognize the scene type (e.g., kitchen, beach), people's expressions and poses, and properties of objects (e.g., the color of objects, their texture). Your task is to stump this smart robot! In particular, it already knows answers to some questions about this scene. We will tell you what these questions are.

Ask a question about this scene that this SMART robot probably can not answer, but any human can easily answer while looking at the scene in the image.
IMPORTANT: The question should be about this scene. That is, the human should need the image to be able to answer the question – the human should not be able to answer the question without looking at the image.



Your work will get rejected if you do not follow the instructions below:

- Do not ask questions that are similar to the ones listed below each image. As mentioned, the robot already knows the answers to those questions for the scene in this image. Please ask about something different.
- Do not repeat questions. Do not ask the same questions or the same questions with minor variations over and over again across images. Think of a new question each time specific to the scene in each image.
- Each question should be a single question. Do not ask questions that have multiple parts or multiple sub-questions in them.
- Do not ask generic questions that can be asked of many other scenes. Ask questions specific to the scene in each image.

Below is a list of questions the smart robot can already answer. Please ask a different question about this scene that a human can answer *if* looking at the scene in the image (and not otherwise), but would stump this smart robot:

Q1: What is unusual about this mustache? (The robot already knows the answer to this question.)

Q2: What is her facial expression? (The robot already knows the answer to this question.)

Q3: Write your question, different from the questions above, here to stump this smart robot.

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Answer Generation:

Help Us Answer Questions About Images!

Updated instructions: Please read carefully

Hide Show

Please answer some questions about images **with brief answers**. Your answers should be how most other people would answer the questions. If the question doesn't make sense, please try your best to answer it and indicate via the buttons that you are unsure of your response.

If you don't follow the following instructions, your work will be rejected.



Your work will get rejected if you do not follow the instructions below:

- Answer the question based on what is going on in the scene depicted in the image.
- Your answer should be a brief phrase (not a complete sentence).
  - "It is a kitchen." -> "kitchen"
- For yes/no questions, please just say yes/no.
  - "You bet it is!" -> "yes"
- For numerical answers, please use digits.
  - "Ten." -> "10"
- If you need to speculate (e.g., "What just happened?"), provide an answer that most people would agree on.
- If you don't know the answer (e.g., specific dog breed), provide your best guess.
- Respond matter-of-factly and avoid using conversational language or inserting your opinion.

10 answers collected from 10 crowdworkers

Please answer the question using as few words as possible:

Q1: What is unusual about this mustache?

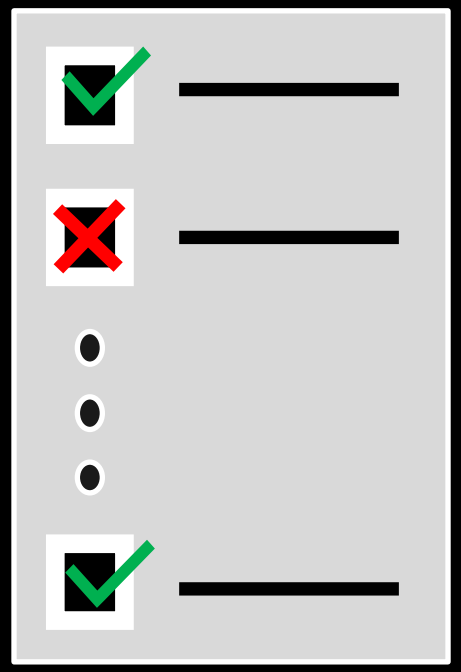
A1: Write your answer here.

Do you think you were able to answer the question correctly?
(Clicking an option will take you to the next question.)

no maybe yes **Page 1/2**

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Approach: Large-Scale Dataset Challenges



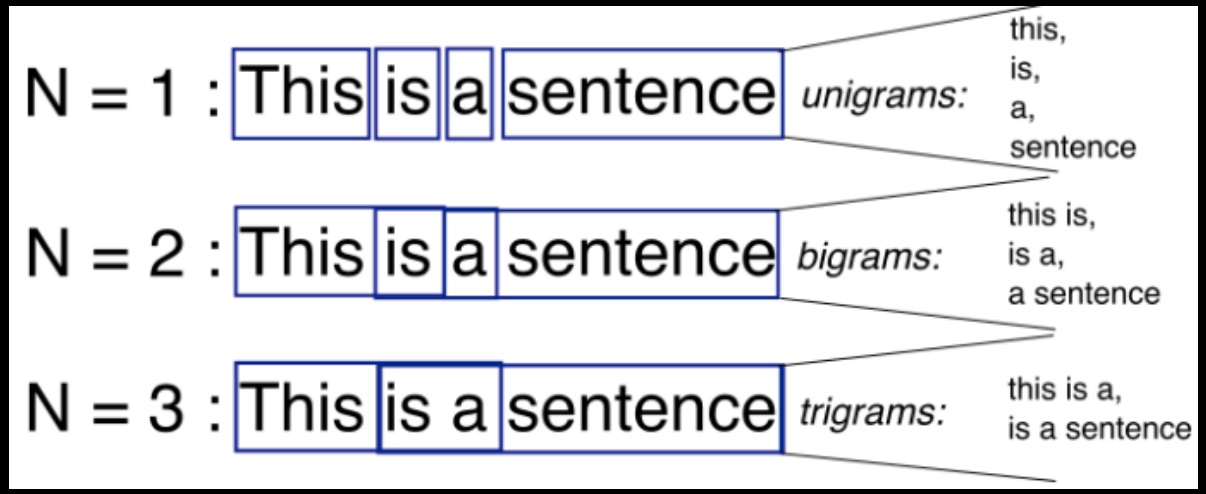
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Image Captioning: Inherit Language Translation and Summarization Metrics

- BLEU (precision–based)
- METEOR
- Rouge (recall–based)

Idea: compute similarities of n-grams between a predicted caption and each ground truth caption



http://recognize-speech.com/language-model/n-gram-model/comparison

- Note: all these metrics are predicated on ONE ground truth

Image Captioning: New CIDEr Metric

Idea: measure similarity of a predicted caption to how most people describe an image based on *n*-grams unique to the image



From Devi Parikh's team



A cow is standing in a field.

A cow with horns and long hair covering its face stands in a field.

A cow with hair over its eyes stands in a field.

This horned creature is getting his picture taken.

A furry animal with horns roams on the range.

Vedantam et al. CIDEr: Consensus-based Image Description Evaluation. CVPR 2015.

VQA: New Accuracy Metric

Idea: measure similarity of a predicted answer to how many people answer the question ("using as few words as possible")



From Devi Parikh's team

$$\text{accuracy} = \min\left(\frac{\#\text{ humans that provided that answer}}{3}, 1\right)$$

Vedantam et al. CIDEr: Consensus-based Image Description Evaluation. CVPR 2015.

VQA: New Accuracy Metric



Does this picture look scary?

(1) yes
(2) no
(3) no
(4) yes
(5) no
(6) yes
(7) yes
(8) no
(9) no
(10) no

What is the accuracy of an algorithm prediction of

- "yes"?
- "no"?
- "maybe"?

$$\text{accuracy} = \min\left(\frac{\#\text{ humans that provided that answer}}{3}, 1\right)$$

VQA: New Accuracy Metric



Which side of the room is the toilet on?

(1) right
(2) left
(3) right
(4) right
(5) right
(6) right
(7) right side
(8) right
(9) center
(10) right

What is the accuracy of an algorithm prediction of

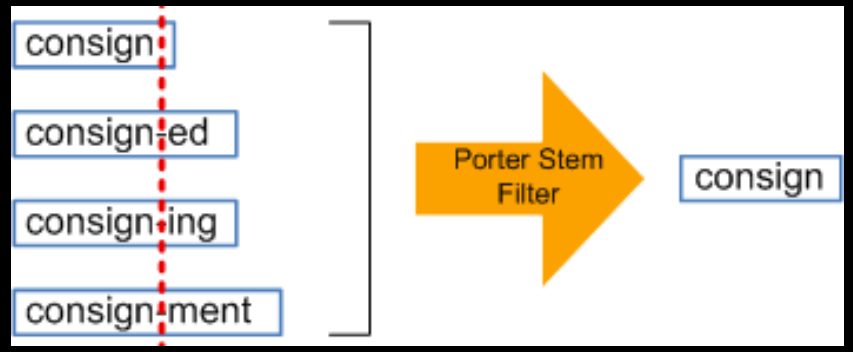
- "right"?
- "left"?
- "right side"?
- "center"?
- "bottom"?

$$\text{accuracy} = \min\left(\frac{\#\text{ humans that provided that answer}}{3}, 1\right)$$

Implementation detail: for fair comparison to humans, there are 10 rounds of comparing a prediction with each possible set of 9 human-supplied answers

Evaluation Implementation Detail: Common to Preprocess Text

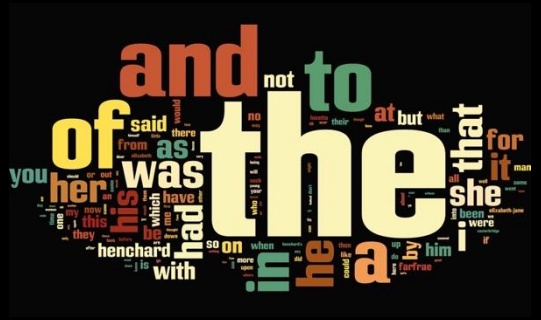
- Lower case all letters
- Use each word's "stem"; e.g., singular vs plural, reconcile different verb forms
  - e.g., stemming



https://dzone.com/articles/using-lucene-grails

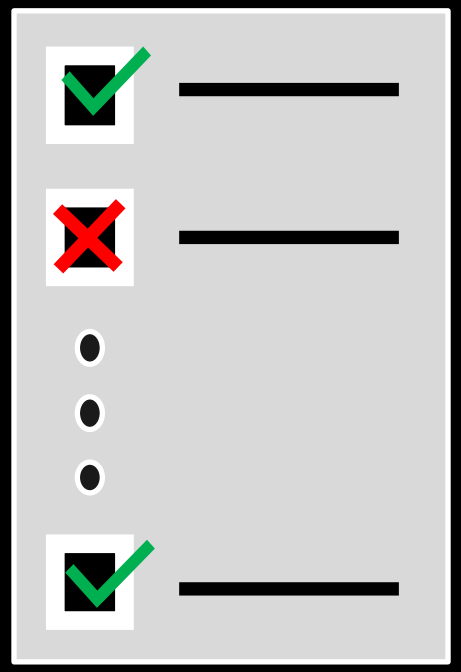
  - e.g., lemmatizing to get intended based word (e.g., caring -> care, not "car")

- Stop word removal: discard frequent words



https://github.com/topics/stopwords-removal

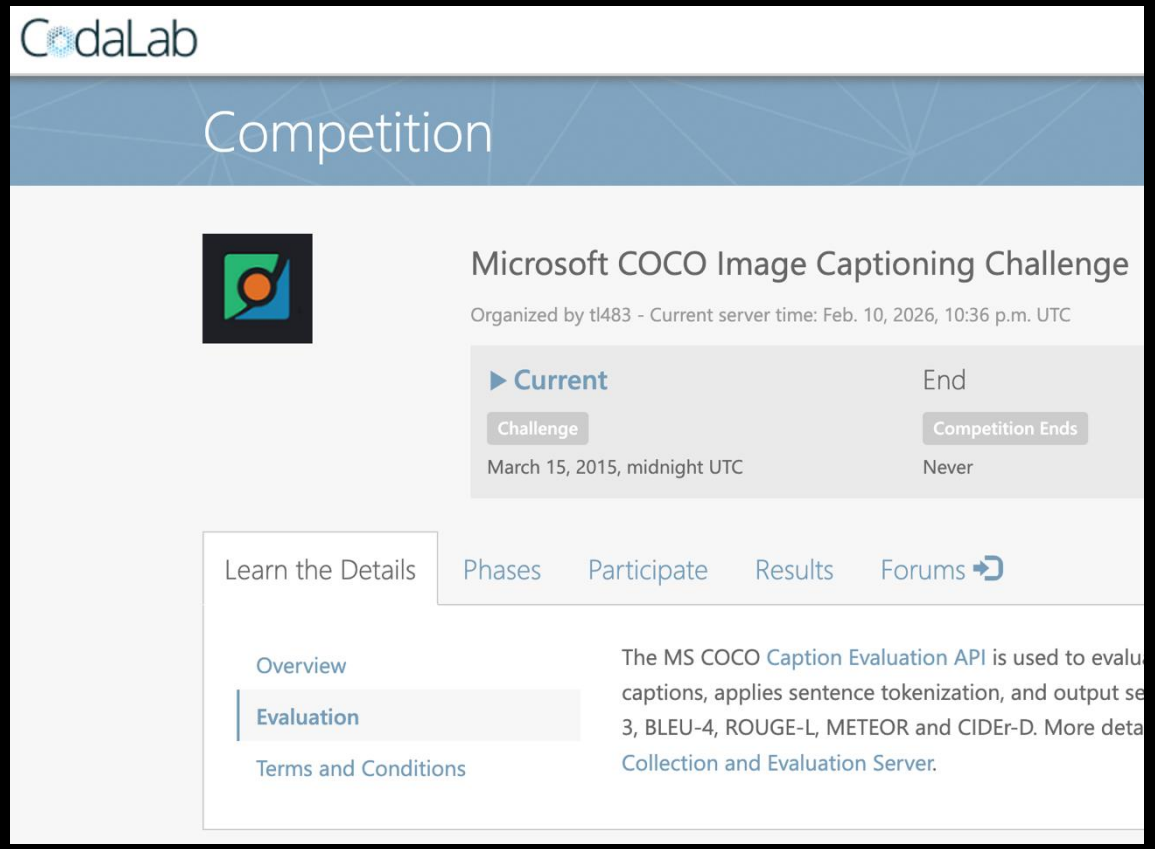
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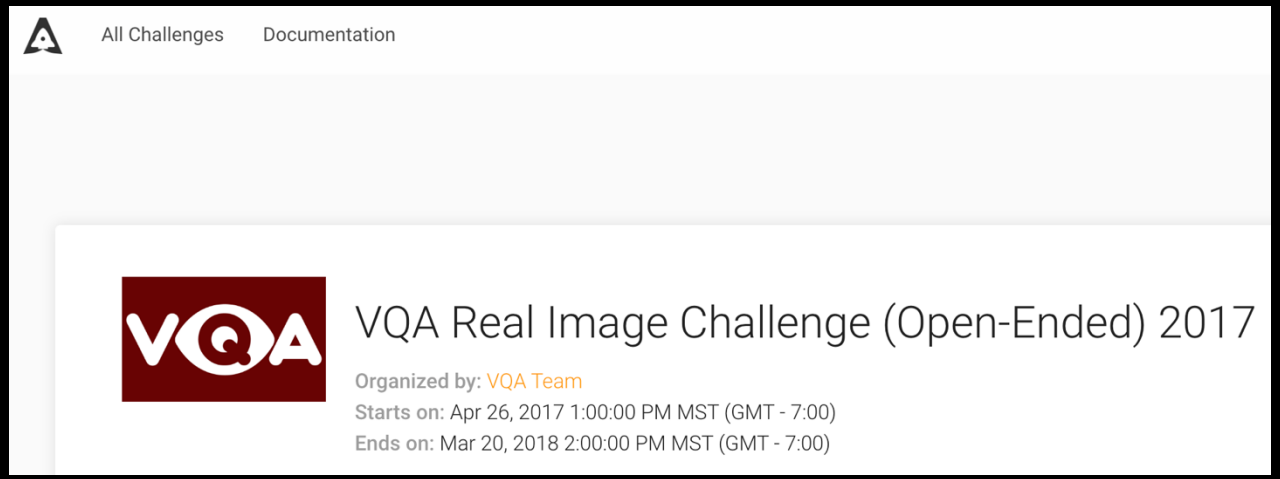
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Image Captioning



VQA





From Devi Parikh's team (EvalAI)

New infrastructure was necessary for the novel tasks with novel evaluation metrics

(defines submission formats for predictions)

Winner Announcements Shared at an Annual Workshop (for 6 Years)

VQA Visual Question Answering

VirginiaTech Invent the Future

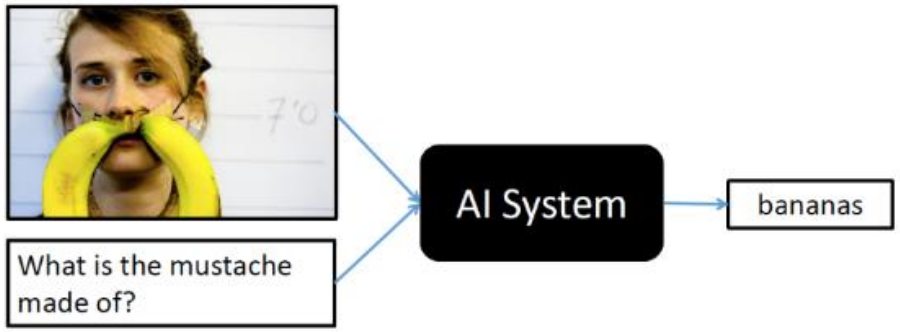
Georgia Tech

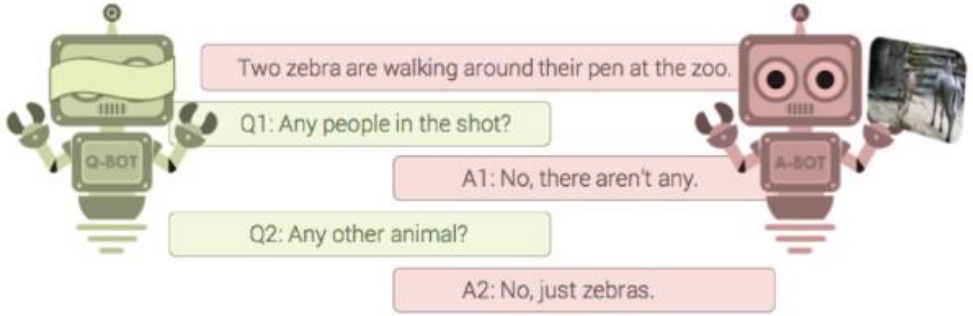
Home | People | Code | Demo | Download | Evaluation | Challenge | Browse | Visualize | Workshop | Sponsors | Terms | External

Visual Question Answering and Dialog Workshop

Location: **Seaside Ballroom B, Long Beach Convention & Entertainment Center**

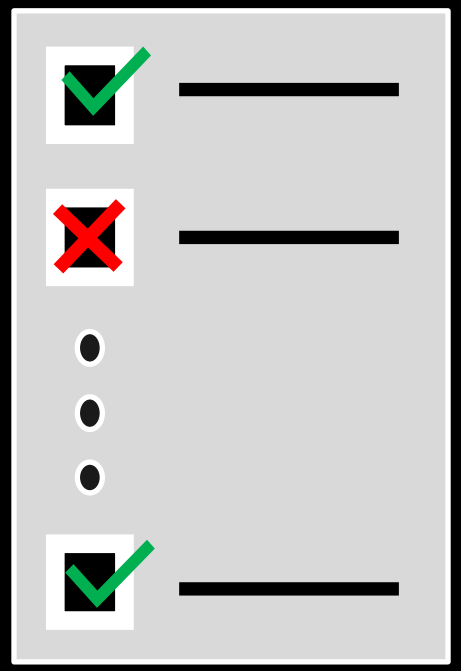
at CVPR 2019, June 17, Long Beach, California, USA





https://visualqa.org/workshop.html

Approach: Large-Scale Dataset Challenges



Ingredients for each challenge:

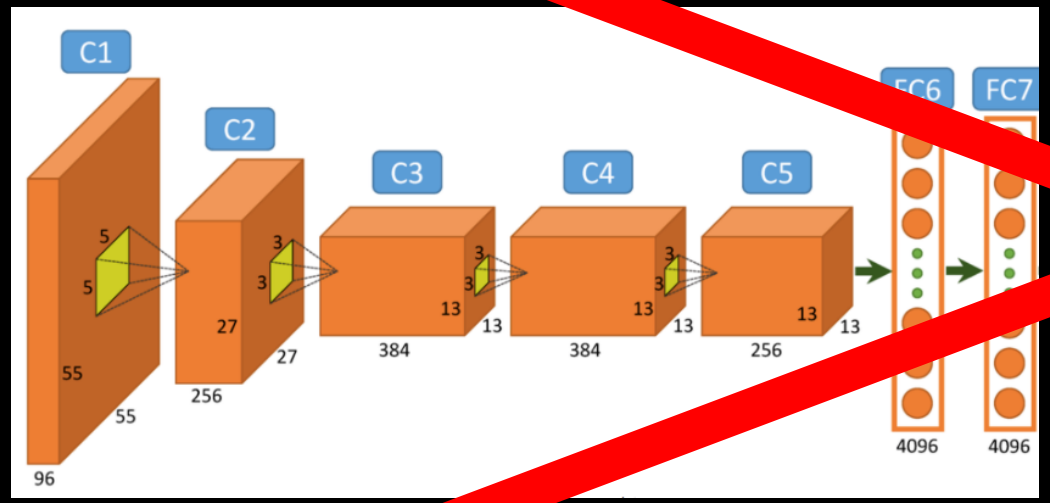
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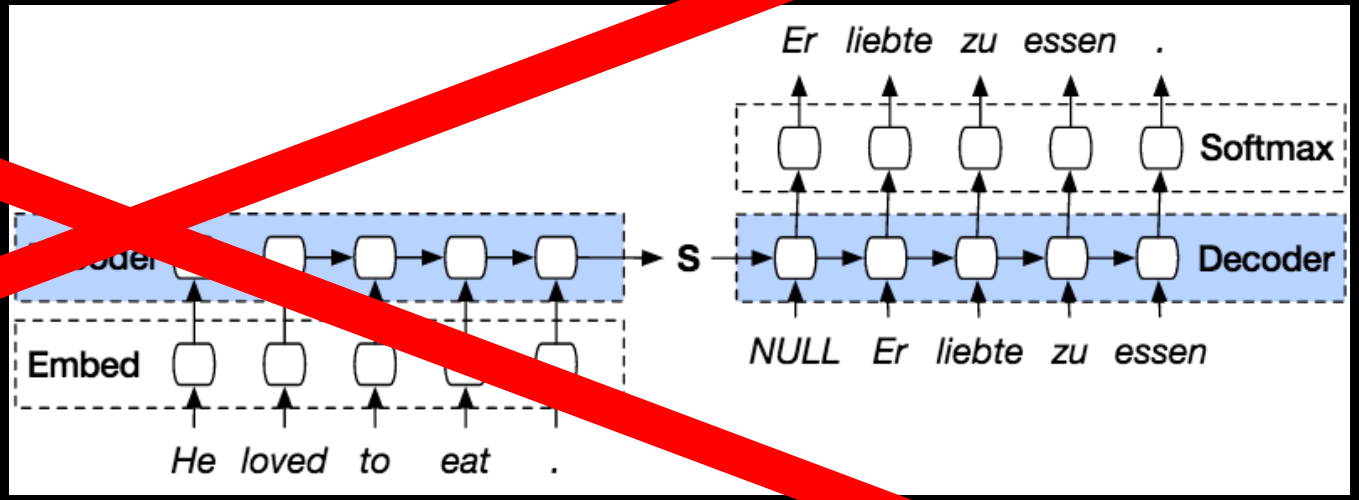
Challenge: Two Modalities in One Framework

e.g., visio[illegible]epresentation with AlexNet



https://www.researchgat[illegible]/figure/Architecture-of-Alexnet-From-left-to-right-inpu[illegible]utput-five-convolutional-layers_fig2_312303454

e.g., language representation with R[illegible]Ns



https://smerity.com/articles/2016/google_nmt_a[illegible]ml

(CVPR, 2015)

Show and Tell: A Neural Image Caption Generator

Oriol Vinyals
Google

Alexander Toshev
Google

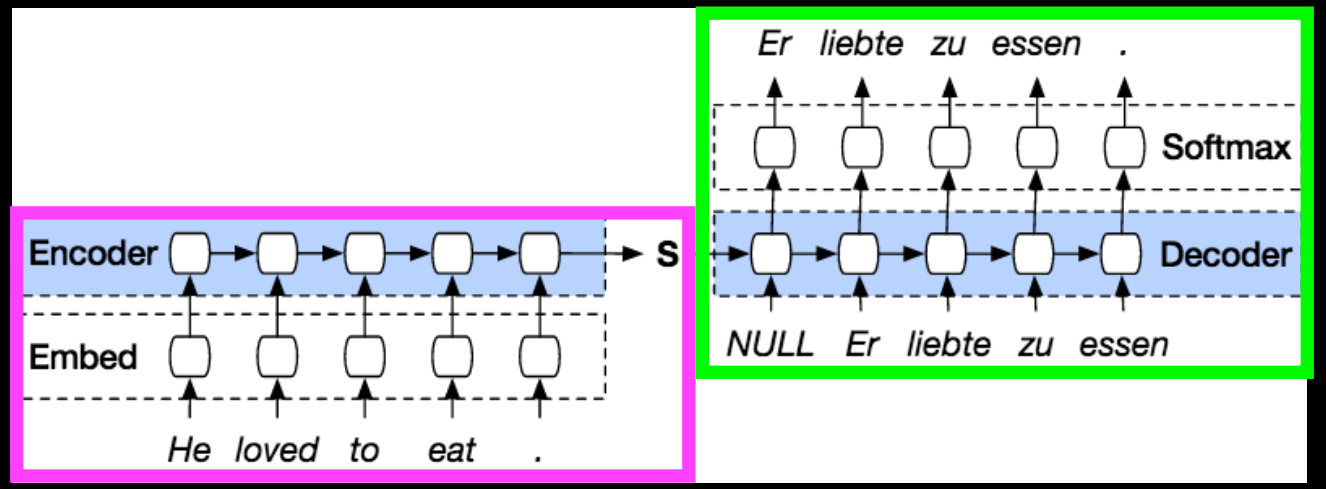
Samy Bengio
Google

Dumitru Erhan
Google

"we would like to present in this work a single joint model…",
rather than the status quo that developers "stitch together"
multiple components in an engineered solution

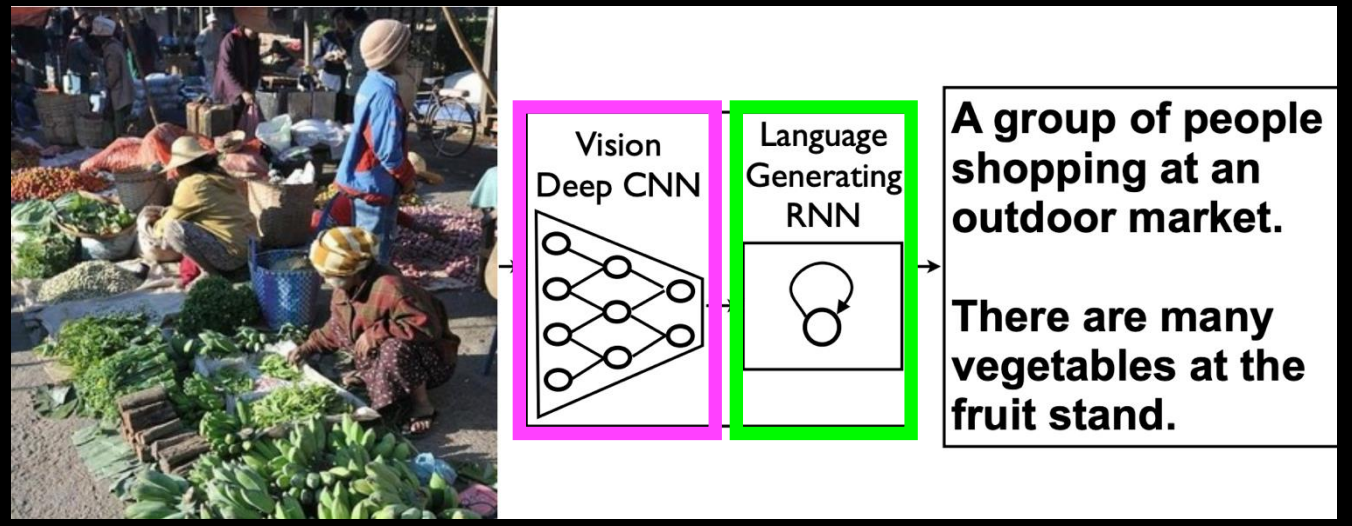
Neural Image Caption Model

Inspiration is machine translation:



https://smerity.com/articles/2016/google_nmt_arch.html

Idea: translate image to text in an end-to-end trained model:

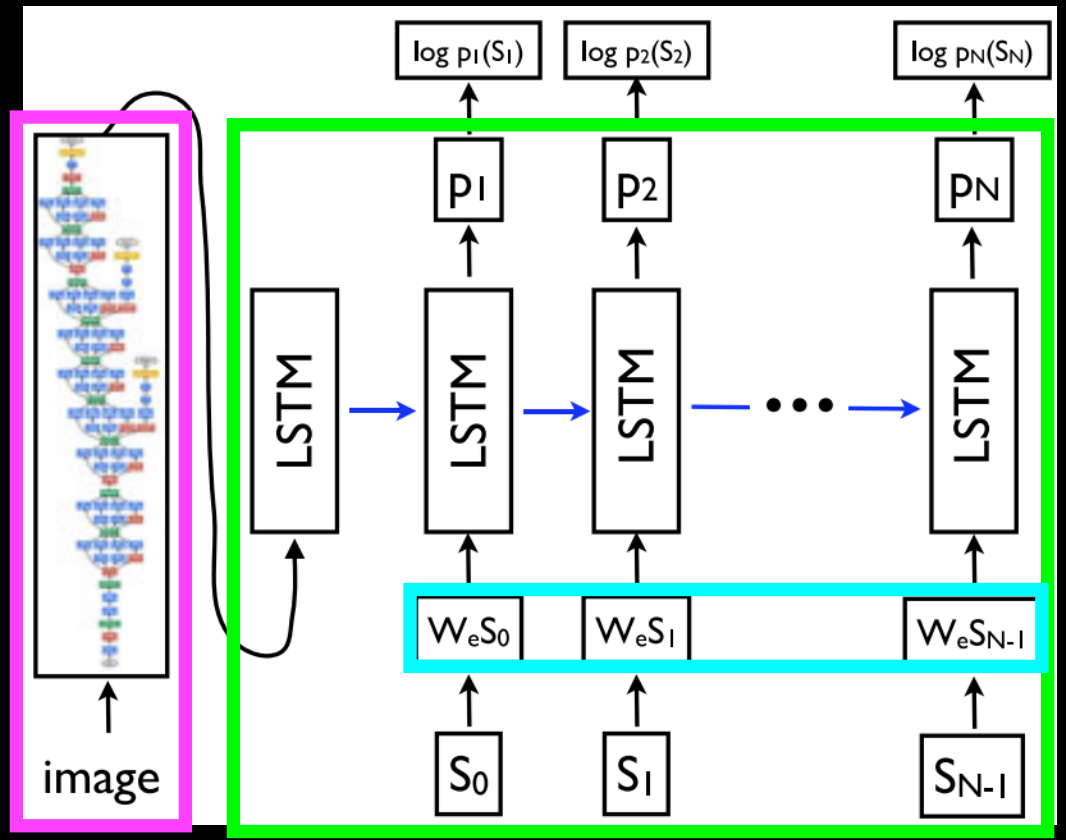


Vinyals et al. Show and Tell: A Neural Image Caption Generator. CVPR 2015.

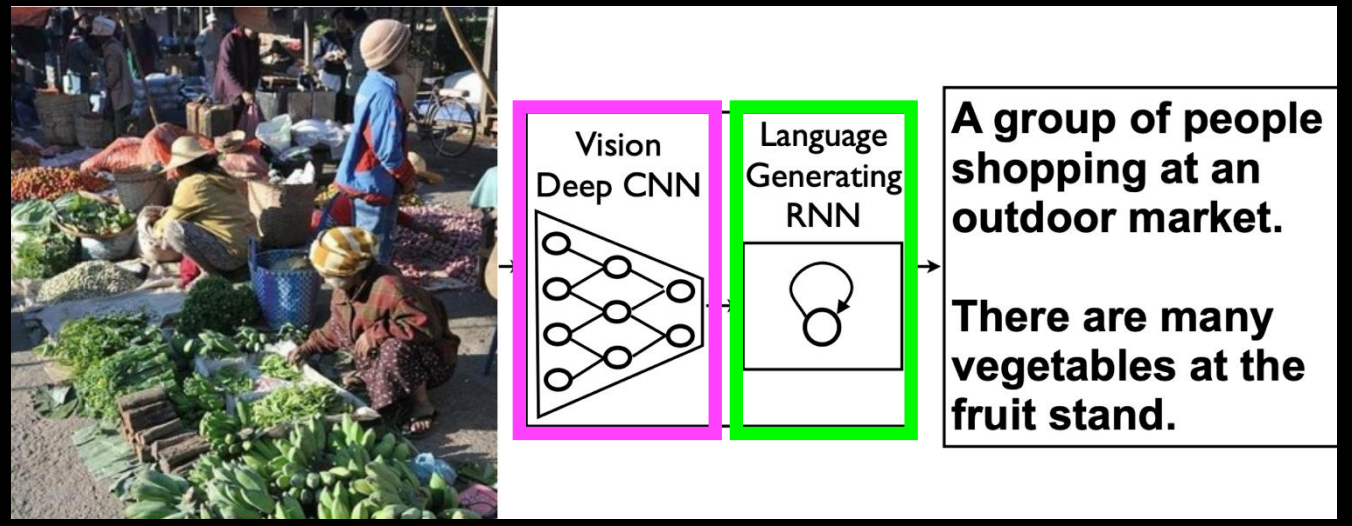
Neural Image Caption Model

2014 ImageNet winner's CNN model

LSTM, predicts next word



* Embedding matrix learned to map words to the same 512-d space as images (for tokenized words seen 5+ times in the training data)

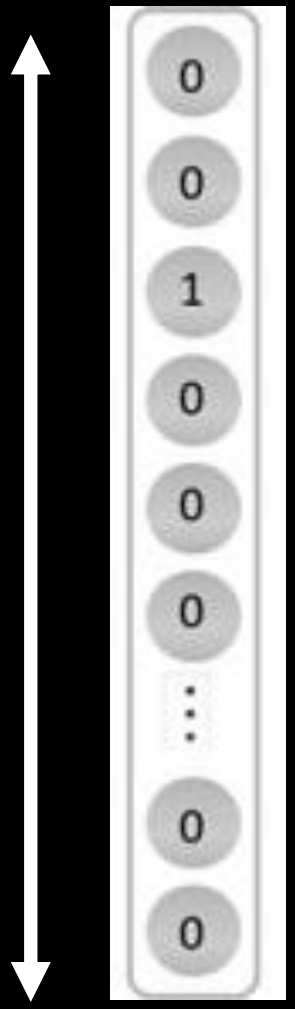


Vinyals et al. Show and Tell: A Neural Image Caption Generator. CVPR 2015.

Word Embedding Motivation (1 of 2)

Dimensionality = vocabulary size

e.g., English alone has ~170,000 words with ~10,000 commonly used words



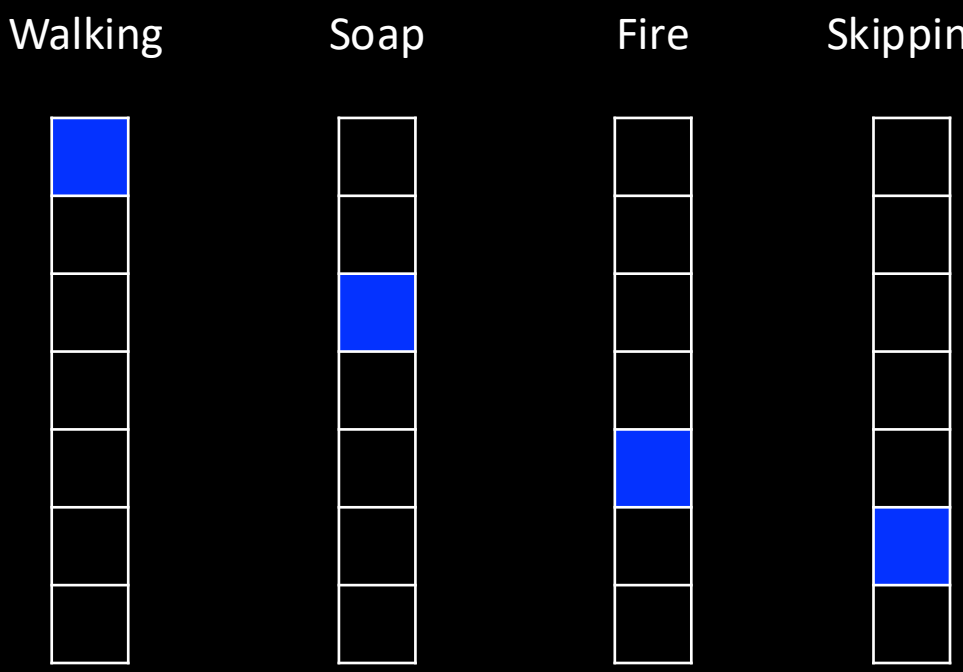
What are problems with one-hot encoding words?

- Huge memory burden
- Computationally expensive

Kamath, Liu, and Whitaker. Deep Learning for NLP and Speech Recognition. 2019

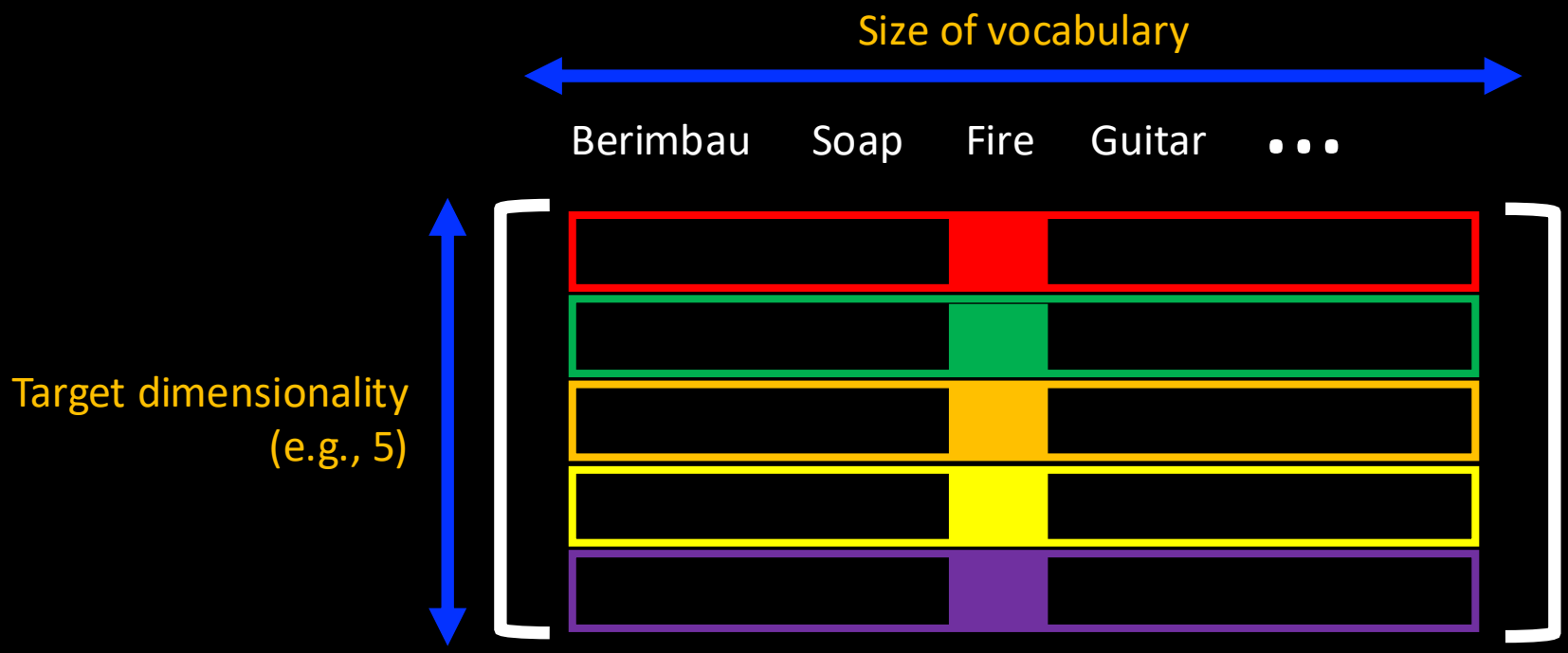
Word Embedding Motivation (2 of 2)

- Also, no notion of which words are similar; the distance between all words is equal!
  - e.g., Shouldn't words that would fit into this context be more similar: "He was ____ to school."

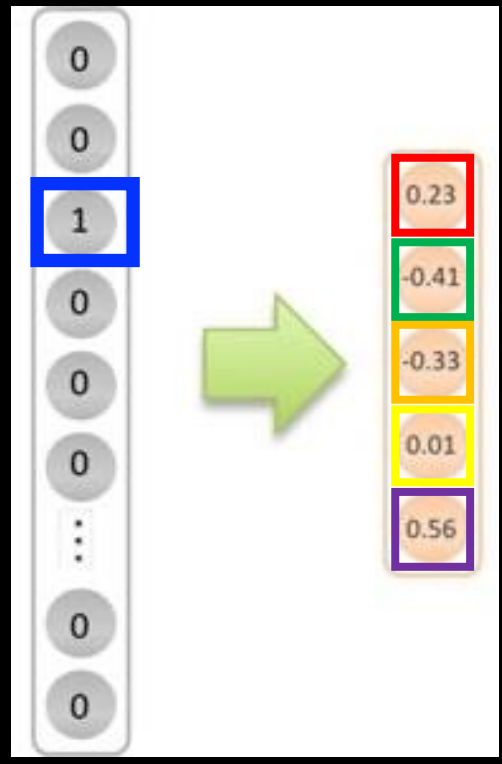


Word Embedding Matrix Implementation

- Converts input word into a dense vector



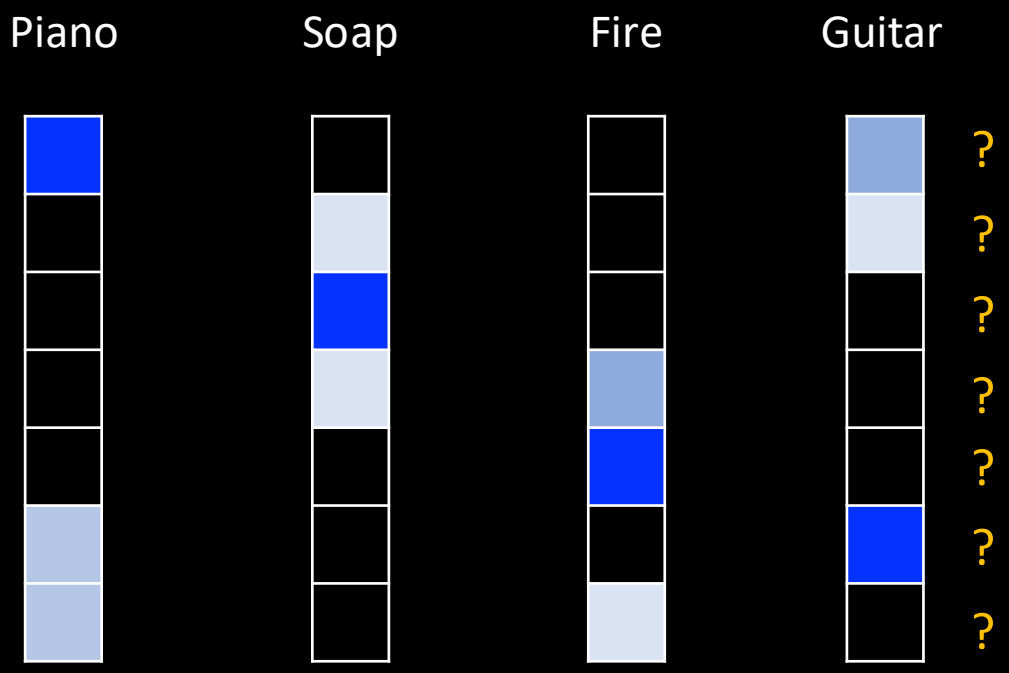
One hot encoding indicates word embedding to use
- efficiently extracted using the one-hot word's index



Kamath, Liu, and Whitaker. Deep Learning for NLP and Speech Recognition. 2019

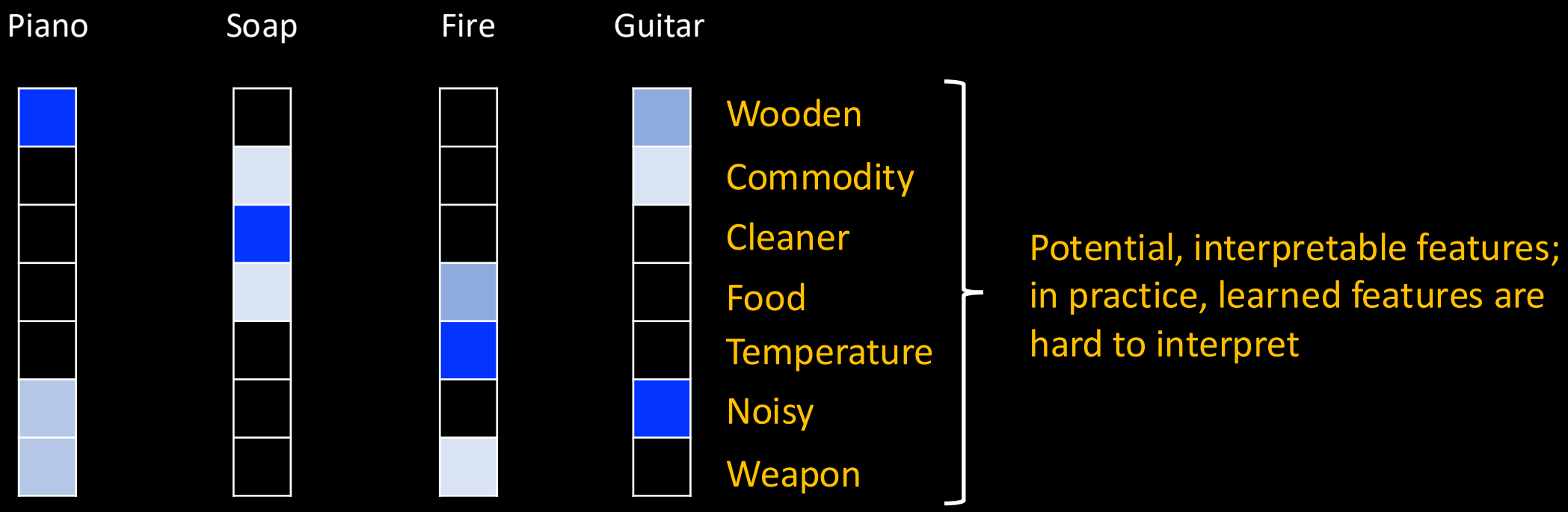
Word Embedding Results

- Dense shared embedding space enables comparing words with a few features, such that the distance between each pair of words differs!
  - What features would discriminate these words?



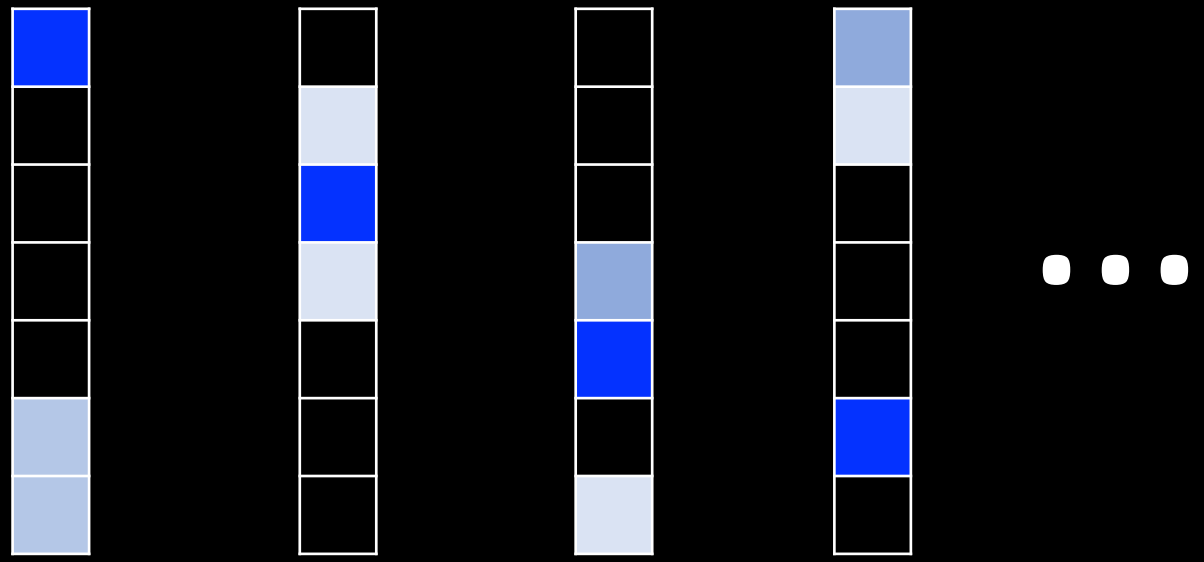
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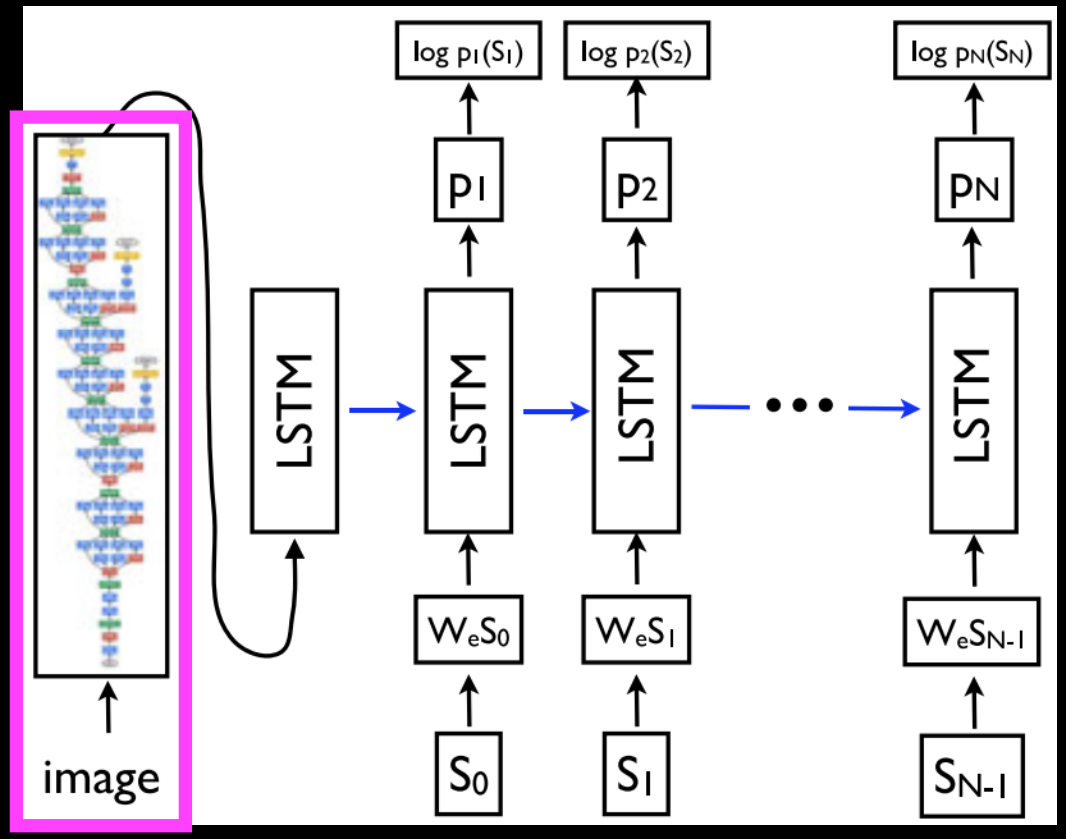


Word Embedding Summary

- An **embedding space** represents a finite number of words, decided in training
- A **word embedding** is represented as a vector indicating its context
- The dimensionality of all word embeddings in an embedding space match
  - What is the word embedding dimensionality for the shown example?



Neural Image Caption Model: Training



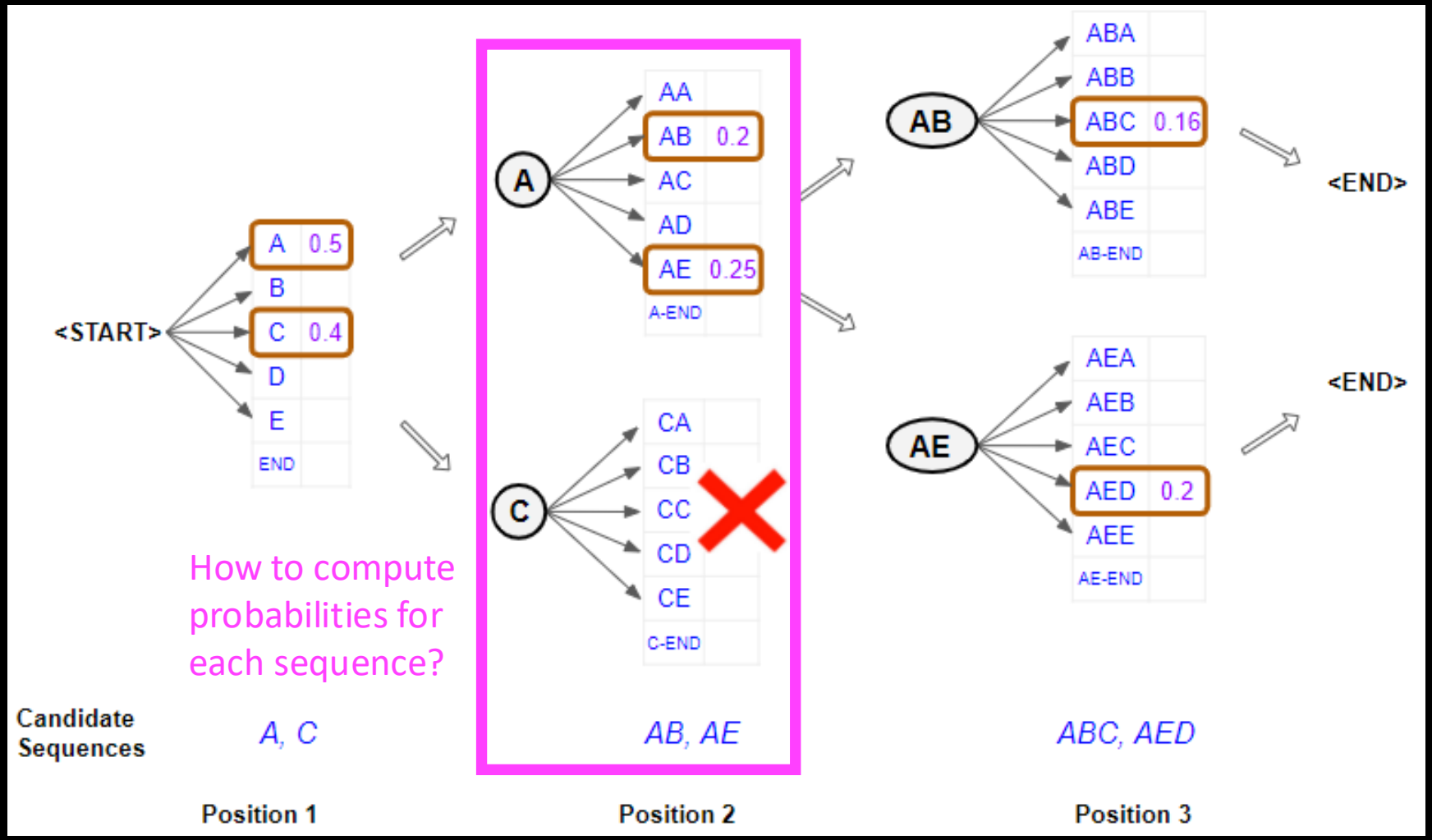
- CNN parameters frozen for 500K steps, and then all parameters trained for 100K steps
- Training took over 3 weeks on a K20 GPU
- Datasets used for model development:

Dataset name	size		
	train	valid.	test
Pascal VOC 2008 [2]	-	-	1,000
Flickr8k [42]	6,000	1,000	1,000
Flickr30k [43]	28,000	1,000	1,000
MSCOCO [44]	82,783	40,504	40,775
SBU [18]	1M	-	-

Vinyals et al. Show and Tell: A Neural Image Caption Generator. CVPR 2015.

Neural Image Caption Model: Used Beam Search

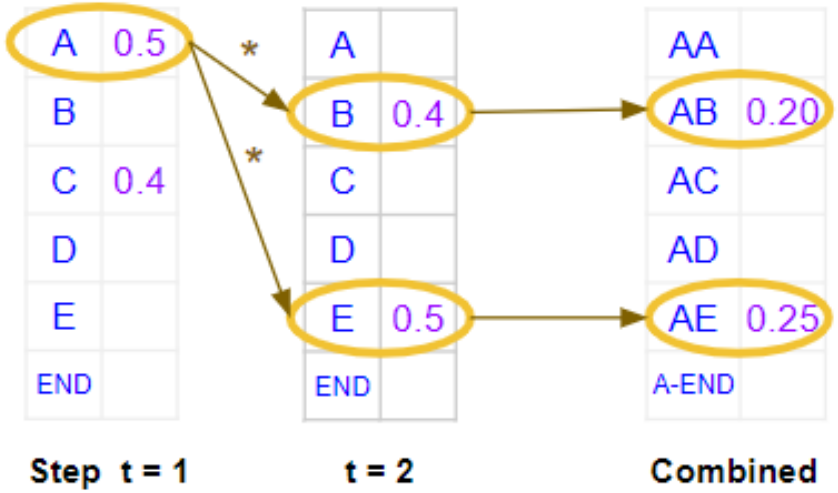
- Default: greedily take best word prediction at each time step
- Idea: choose most probable from top predictions of current plus preceding words (length N)
- Beam width: number of top predictions to keep at each step
- In the example:
  - What is the beam width?
  - How many times is the model run after each prediction?
- For NIC model, beam width=20



https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24/

Neural Image Caption Model: Beam Search

Softmax outputs:



Prob (AB | input) = Prob (A | input) * Prob (B | A, input)

Prob (AB) = Prob (A) * Prob (B | A)

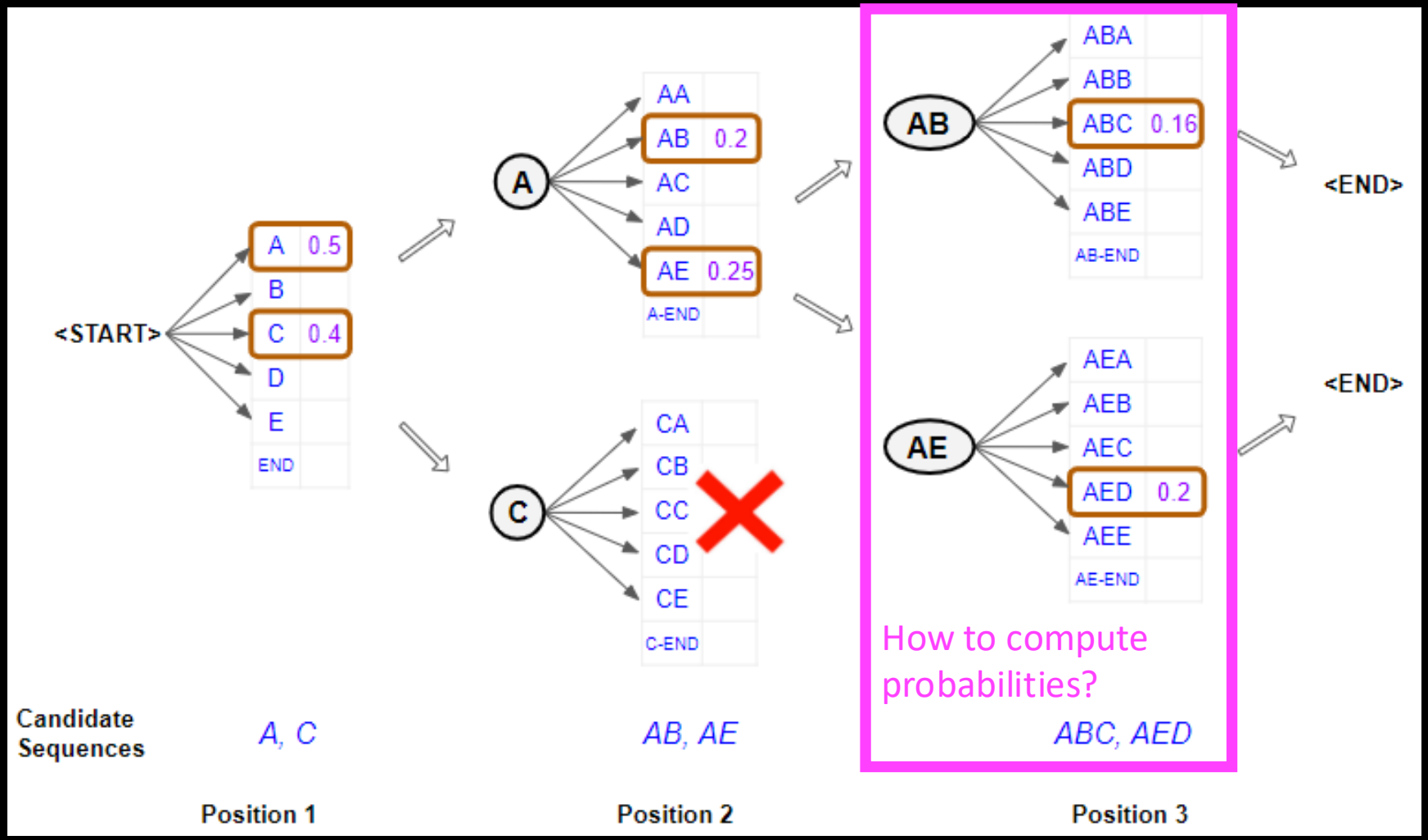
= 0.5 * 0.4

= 0.20

https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24/

Neural Image Caption Model: Beam Search

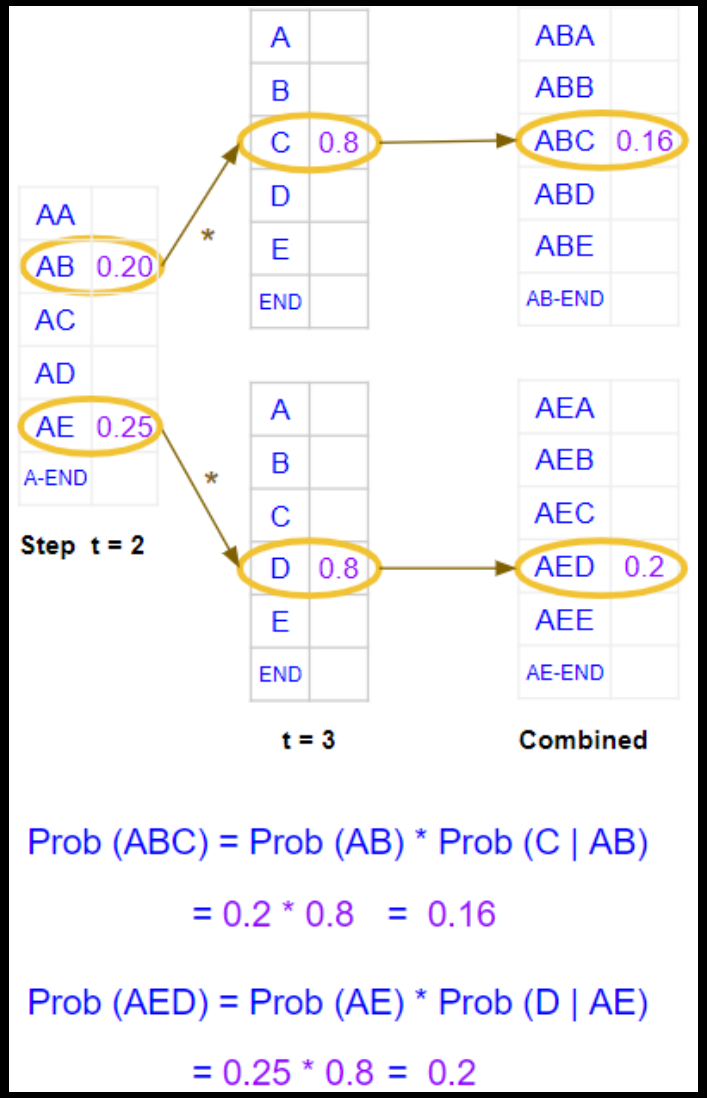
- Default: greedily take best word prediction at each time step
- Idea: choose most probable from top predictions of current plus preceding words (length N)
- Beam width: number of top predictions to keep at each step



https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24/

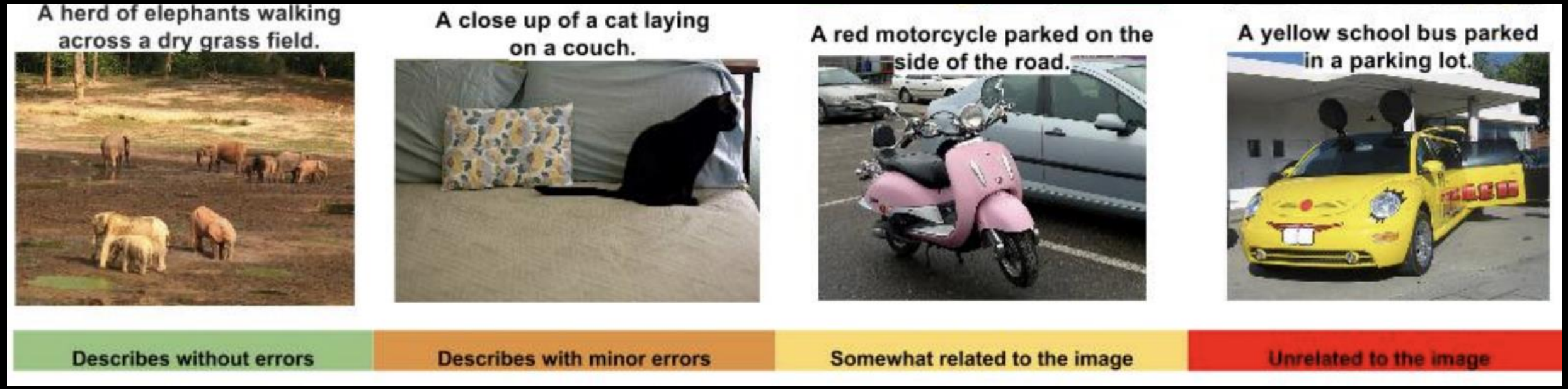
Neural Image Caption Model: Beam Search

Softmax outputs:



https://towardsdatascience.com/foundations-of-nlp-explained-visually-beam-search-how-it-works-1586b9849a24/

Neural Image Caption Model: Results



Achieved state-of-the-art, yet imperfect results; hypothesis is more training data would boost performance

Nearest neighbor words in learned embedding space had similar words close to each other:

Word	Neighbors
car	van, cab, suv, vehicule, jeep
boy	toddler, gentleman, daughter, son
street	road, streets, highway, freeway
horse	pony, donkey, pig, goat, mule
computer	computers, pc, crt, chip, compute

Vinyals et al. Show and Tell: A Neural Image Caption Generator. CVPR 2015.

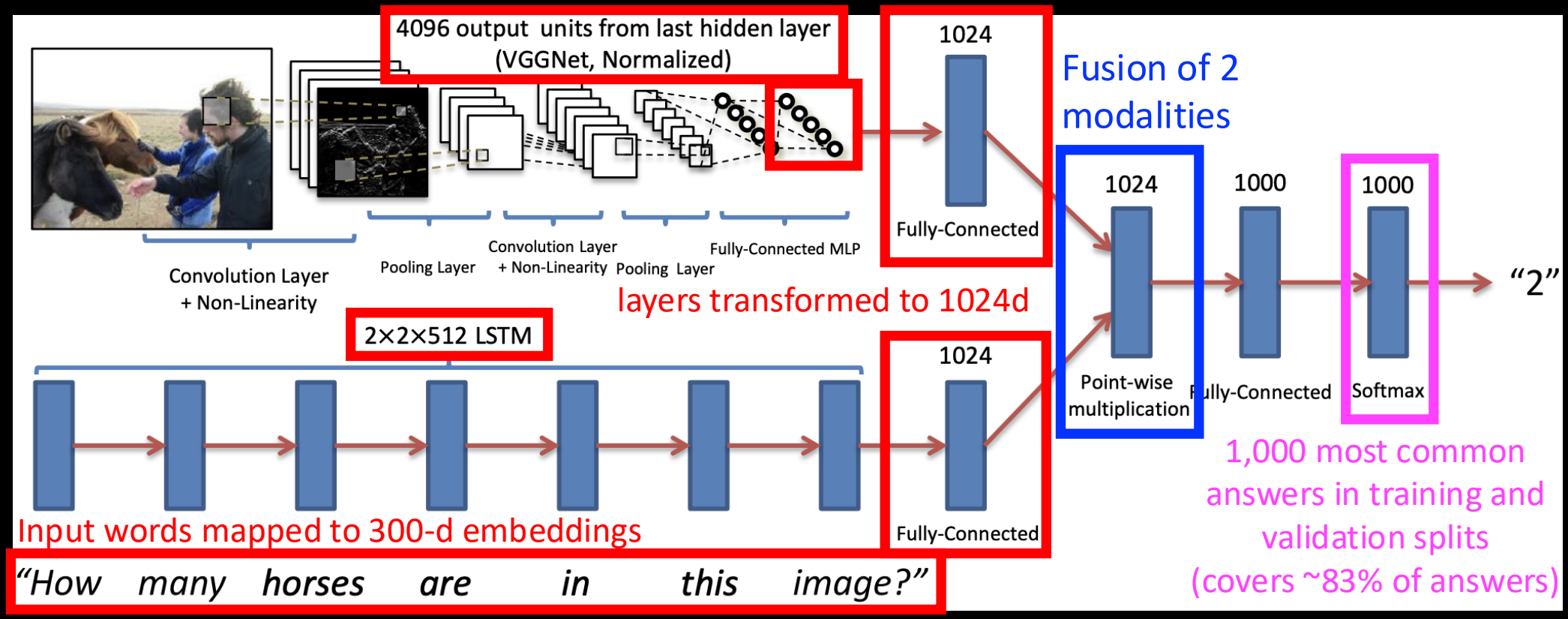
(ICCV, 2015)

VQA: Visual Question Answering

www.visualqa.org

Aishwarya Agrawal*, Jiasen Lu*, Stanislaw Antol*,
Margaret Mitchell, C. Lawrence Zitnick, Dhruv Batra, Devi Parikh

Proposed Model



Trained end-to-end with cross-entropy loss, freezing VGGNet parameters

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Model Analysis

Baseline: most popular answer from train/val splits

Baseline: 1-hidden layer LSTM and image activations without normalization

Proposed model

	All
prior ("yes")	29.66
LSTM Q + I	53.74
deeper LSTM Q + norm I	**57.75**

How does the model's performance compare to the simpler baselines?

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Model Analysis

Baseline: most popular answer from train/val splits

Baseline: 1-hidden layer LSTM and image activations without normalization

Proposed model

	All	Yes/No	Number	Other
prior ("yes")	29.66	70.81	00.39	01.15
LSTM Q + I	53.74	78.94	35.24	36.42
deeper LSTM Q + norm I	**57.75**	**80.50**	**36.77**	**43.08**

What trends are observed across different question types?

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

Model Analysis

	All	Yes/No	Number	Other
prior ("yes")	29.66	70.81	00.39	01.15
Only image: I	28.13	64.01	00.42	03.77
Only question: LSTM Q	48.76	78.20	35.68	26.59
LSTM Q + I	53.74	78.94	35.24	36.42
Only question: deeper LSTM Q	50.39	78.41	34.68	30.03
deeper LSTM Q + norm I	**57.75**	**80.50**	**36.77**	**43.08**

Which modality is most predictive?

How does each modality, vision and language, influence performance?

Agrawal et al. VQA: Visual Question Answering. ICCV 2015.

The Pioneering Multimodal Work Set Blueprint for Many New Challenges, Including from My Team

(Evaluation Metrics, Evaluation Servers, Annotation Systems, Annual Workshop)



Caption:
A bunch of small light brown mushrooms in a green field.

Answer Visual Question:
Q: Is it edible or poisonous?

A: Poisonous

VizWiz: first publicly-available datasets originating from authentic use cases

(extended 44,799 submissions from 11,045 people seeking visual assistance in 2011-2015)

VQA Impact Celebrated 10 Years Later



Today's Topics

- Isolated successes: CNNs for vision and RNNs for language
- The next challenge: describing visual scenes with natural language
- Early vision-language pioneers: connecting pixels and words
- Programming tutorial

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The End